Analisis AXIS ID5_PM14

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1) Identificación del artículo

Referencia: Chilwan U., Shameer S., Hanan A., Balasubramanium R.K. Effect of 6-Min Mastication Test on Masticatory Function in Young Versus Older Adults: A Comparative SEMG Study. The Scientific World Journal. 2025; Article ID 1506278. DOI: 10.1155/tswj/1506278. (Received 21 Feb 2025; Accepted 21 Aug 2025.)

Aprobación/Financiación/COI: Ethics approval IEC KMC MLR-11/2022/452 (17-11-2022) (p.6). Funding: Indian Council of Medical Research (Grant No.: IIRP23/0256) (p.6). Authors declare no conflicts (p.6). Data available on request (p.6).

2) Resumen rápido (contexto)

Design: Cross-sectional comparative study (n=50; 25 young 18–35 y 25 older ≥60). Single 6-minute mastication test (6MMT) chewing two gum pieces at 80 bpm; SEMG recorded from bilateral masseter and submental regions using Chattanooga VitalStim Plus (therapeutic device). Outcomes: mean amplitude, peak amplitude, MVC (%), and power computed per muscle; within- and between-group comparisons (paired t, independent t), effect sizes reported. (Methods pp.2–4; Tables 1–5.)

Main outcomes: Older adults had significantly reduced masseter parameters (mean, peak, MVC, power) with large effect sizes; submental showed difference only in peak value between groups; within-group fatigue effects on submental mean and power across 1st vs 6th minute in both groups. (Results pp.4–5; Tables 2–5.)

Data available: summary statistics and effect sizes in tables; raw device data export to Excel; datasets available from corresponding author on reasonable request (p.6). **Note:** device technical specs (sampling rate etc.) not disclosed by manufacturer (Methods p.3; Limitations p.5).

3) AXIS ítem por ítem

1) ¿Objetivos claros?

Evalúa: si objetivos/hypótesis explícitos.

Juicio: Yes.

Evidencia: “Therefore, this study is aimed at assessing and comparing the parameters of masticatory function in the masseter and submental muscles across young and older adults using SEMG. We hypothesized that older adults would exhibit significantly reduced masticatory muscle performance and endurance…” (Intro pp.1–2).

2) ¿Diseño apropiado?

Evalúa: la adecuación del diseño al objetivo.

Juicio: Yes.

Evidencia: Cross-sectional comparison of SEMG during 6MMT between age groups is appropriate to detect age differences in endurance/fatigue (Study Design, Methods 2.1–2.4; pp.2–4).

3) ¿Justificación del tamaño muestral?

Evalúa: a priori sample size/power.

Juicio: No.

Evidencia: Authors state no a priori calculation; sample chosen by feasibility (n=25/group) (Sample Size 2.5 p.3). They report effect sizes and CIs afterwards but no power justification.

4) Población objetivo definida?

Evalúa: claridad de población and criteria.

Juicio: Yes.

Evidencia: Inclusion/exclusion criteria detailed (ages, health, exclusions for TMD, missing teeth, implants, pregnancy etc.) and group counts (n=50) (Participants 2.2; Table2 p.4).

5) Marco muestral adecuado?

Evalúa: fuente del muestreo.

Juicio: Partial.

Evidencia: Participants recruited (site implied Kasturba Medical College) but recruitment method (random/consecutive/volunteer) not fully described (Participants 2.2 p.2–3).

6) ¿La selección produce una muestra representativa?

Evalúa: representatividad.

Juicio: Partial.

Evidencia: Balanced groups n=25 but small convenience sample from single center limits external generalizability (Limitations p.5).

7) ¿No-responders abordados?

Evalúa: tasa de respuesta / comparación.

Juicio: No.

Evidencia: No recruitment flowchart and no info on refusals/non-responders; not reported.

8) ¿Variables/outcomes apropiados?

Evalúa: si measures fit objectives.

Juicio: Yes.

Evidencia: SEMG parameters (mean amplitude, peak, MVC, power) are valid outcomes for masticatory function and endurance (Table1; Methods 2.4–2.6 pp.3–4).

9) ¿Mediciones con instrumentos validados/pilotadas?

Evalúa Instrument validation and measurement detail.

Juicio: Partial.

Evidencia: Device used Chattanooga VitalStim Plus (therapeutic), BUT manufacturer did not disclose key technical specs (sampling rate, filters, input impedance) (Methods p.3). Authors note device choice limitation (Limitations p.5). Electrode placement and skin prep described (Methods 2.4 p.2–3).

10) Criterios de significancia/precisión claros (p, IC)?

Evalúa: thresholds and precision reporting.

Juicio: Partial.

Evidencia: p-values reported and effect sizes (Cohen's d, Hedges' g) with 95% CIs mentioned for effect sizes in some places; no confidence intervals shown for main outcome means; normality tested via Shapiro-Wilk (Statistical Analysis 2.6 p.3).

11) ¿Métodos descritos suficientes para replicar?

Evalúa: reproducibility of protocol.

Juicio: Parcial.

Evidencia: Detailed task protocol (gum, metronome at 80 bpm, timing, electrode placement, posture); signal exported to Excel; BUT key device acquisition parameters undisclosed by the manufacturer, so exact replication and signal quality assessment are limited (Methods 2.4–2.6; Limitations p.5).

12) ¿Datos básicos descritos adecuadamente?

Evalúa: baseline demographics and variables.

Juicio: Yes.

Evidencia: Table 2 descriptive stats and Tables 3–5 show within/between tests with means, medians, SDs, effect sizes (pp.4–5).

13) ¿Tasa de respuesta genera preocupación?

Evalúa: response rate reporting.

Juicio: No (not reported).

Evidencia: No reporting of how many invited/consented/refused; not provided.

14) ¿Info sobre no-responders descrita?

Evalúa: comparison responders vs non-responders.

Juicio: No / Not reported.

15) ¿Resultados internamente consistentes?

Evalúa: coherence between text/tables/figures.

Juicio: Yes.

Evidencia: Text claims align with Tables 2–5 (e.g., masseter differences $p<0.05$ and large d reported Table5 p.5).

16) ¿Resultados presentados para análisis descritos en métodos?

Evalúa: completeness.

Juicio: Yes.

Evidencia: Paired t for 1st vs 6th minute and independent t between groups performed and results shown (Tables 3–5 pp.4–5).

17) ¿Discusión/conclusiones justificadas por resultados?

Evalúa: interpretation appropriate.

Juicio: Yes with caution.

Evidencia: Authors link observed declines to sarcopenia/neural changes but acknowledge device and sample limitations (Discussion & Limitations pp.4–6).

18) ¿Limitaciones discutidas?

Evalúa: acknowledgement of limitations.

Juicio: Yes.

Evidencia: Authors list sample size, cross-sectional design, device technical unknowns, subcutaneous tissue influence, non-standardized bolus as limitations (Limitations p.5).

19) ¿Financiación/COI declarados?

Evalúa: funding and conflicts.

Juicio: Yes.

Evidencia: Funding ICMR grant No IIRP23/0256 (p.6); authors declare no conflicts (p.6).

20) ¿Ética/consentimiento?

Evalúa: IRB approval & informed consent.

Juicio: Yes.

Evidencia: Ethics committee approval IEC KMC MLR-11/2022/452 (17-11-2022) and signed informed consent (p.6).

Juicio general AXIS y justificación ID5_PM14

Reporting quality: Moderate. Strengths — clear objectives, detailed task protocol (6MMT cadence, gum, electrode placement, posture), appropriate statistical tests, effect sizes reported, ethics and funding declared (Methods & Results pp.2–6; Tables).

Main limitations impacting AXIS appraisal: no a priori sample size calculation (item 3), small convenience sample from single center (items 5–6), no

recruitment/response flow (items 7,13,14), use of a therapeutic SEMG device with undisclosed acquisition specs (sampling rate etc.) which limits reproducibility and signal-quality assessment (items 9,11, Limitations p.5). Authors acknowledge many of these limitations.

Overall: results support age-related decline in masseter SEMG parameters and fatigue effects in submental muscle, but findings are exploratory; replication with diagnostic-grade acquisition and larger, representative samples recommended before clinical translation.